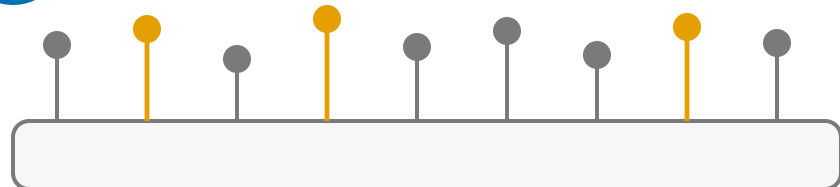


1

Random



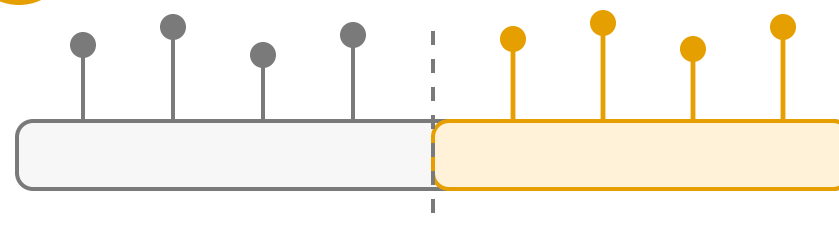
● training variant

● held-out variant

random held-out variants

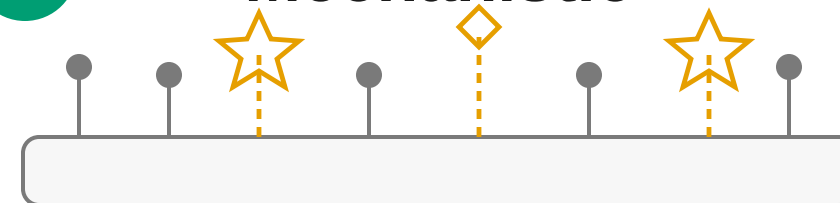
2

Positional

train
N-terminal halfhold out
C-terminal halfhold out one protein region
(C-terminal half)

3

Mechanistic



● other variants

◇ held-out functional sites

hold out hotspot or
functional-switch residues

4

Cross-gene



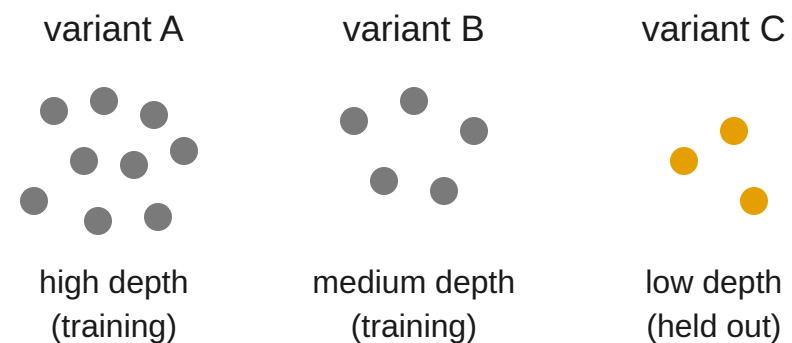
□ training genes

□ held-out gene

hold out a whole gene,
train on the others

5

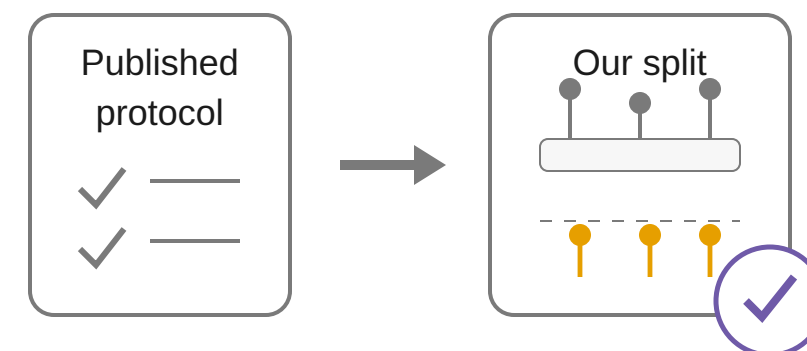
Low-depth

high depth
(training)medium depth
(training)low depth
(held out)

hold out shallow-sampling variants

6

Compatibility



match a prior evaluation protocol